SATISH PUPPALA

Technical Engineering Manager · C/C++ Product Engineering · GenAI and Agentic AI

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SUMMARY

Technical Engineering Manager with **17 years** in C and C++ product development across telecom, retail, gaming and enterprise document management, including **4 years in engineering leadership**. Leads an **11-person team at OpenText** that owns the full SDLC for two enterprise viewer products used by **4,000+ customers**. Stays close to the code: reviews designs, steps in to resolve complex debugging issues when needed, and owns architectural decisions. Over the last two years, moved the team's repetitive work (security fixes, code review, porting and test writing) onto **in-house AI agents**, cutting that effort by **about 25%** and freeing capacity for cross-functional work beyond the team's roadmap. Named inventor on **7 patent applications**.

SKILLS

Leadership: Hiring, **coaching and mentoring**, **performance management**, performance improvement coaching, conflict resolution, **career development**, succession planning, **technical strategy and roadmap**, **delivery and risk management**, technical debt management, **engineering KPIs**, **global teams**, **AI adoption**, process improvement, capacity planning, **Agile/Scrum**, release planning, customer escalations, **stakeholder management**, partnering with product, QA and support

Engineering: C, Modern C++ (C++11/14/17), STL, multithreading and concurrency, memory and performance optimization, **data structures and algorithms**, **system design**, design patterns, OOP, TCP/IP, Java, Python, Win32, COM, MFC, legacy modernization

AI: GitHub Copilot (custom instructions, agents), Claude Code, Cursor, OpenText Aviator, **agentic AI**, MCP, RAG, A2A, Pydantic AI, prompt engineering, transformer-based LLM training and fine-tuning

Debugging and Build: WinDbg, gdb, Valgrind, root cause analysis, **CMake, MSBuild**

Quality, Security and Delivery: GoogleTest, CppUnit, test automation, CI/CD (**Jenkins, GitLab CI**), Fortify, Black Duck, code review practices, SDLC

Cloud and Platforms (working knowledge): Cloud fundamentals, microservices architecture, Docker, Kubernetes, n8n workflow automation

Tools: Visual Studio, Git, Perforce, Mercurial, SVN, ClearCase, Jira, ValueEdge, Windows, Linux

EXPERIENCE

Manager, Software Engineering **Jun 2023 – Present**

OpenText | Hyderabad

- Lead **11 engineers and testers (9 direct reports)** for Windows Viewer (C++) and Web Viewer (Java), serving as both Engineering Manager and Scrum Master. **Hired three engineers and supported three promotions**; own coaching, performance reviews, performance improvement coaching and conflict resolution.
- Led the **32-bit to 64-bit migration** of a 25-year-old desktop viewer with many internal and third-party dependencies (Datalogics, BlackIce, ODMA). **Delivered on schedule in one year** with two developers and two QA engineers, unblocking customers moving to 64-bit SAP.
- Delivered **four annual releases** (23.4, 24.4, 25.4, 26.4), including three 64-bit releases, with **end-to-end SDLC ownership**. Manage approximately 50 customer support cases annually.
- Built **six AI agents** in collaboration with the team, providing technical guidance and coaching: security fix (Fortify and Black Duck), back-porting to older branches, code review, fix verification against acceptance criteria, test automation, and a ValueEdge notifier that posts new issues and support updates to Teams.
- Designed the agents to plug into tools the team already uses (Fortify, Black Duck, ValueEdge, Git, Teams) rather than a new platform, with **an engineer approving every change** before merge.
- Security findings dropped from **about 50 to 30 a year**. The time saved frees **roughly 25% of team capacity** for cross-team initiatives, including LibreOffice contributions and support for the Intelligent Viewer team.
- Took language-pack testing from **0 to 100% automated**. Replaced a nonfunctional Visual Basic test suite with a new framework that already covers **60% of functional tests**.
- Wrote GitHub Copilot customization standards (Guide, Complete Reference), now **adopted by all 4 teams** in the group. Run AI Power Hour for the business unit, and ran a C++ Think Tank every two weeks for 5 months with about 15 developers.

Lead Software Engineer **Sep 2021 – Jun 2023**
OpenText | Hyderabad

- Designed and built modules for OpenText Imaging Windows Viewer, used for document viewing, markup and distribution in large enterprises.
- Owned support tickets and security fixes, led code reviews, and helped the team deliver committed features; stood in as Scrum Master. **Promoted to Manager in June 2023.**
- Received the **High Five Award (Q3 2022)** for the Viewer 22.4 release, language-pack deliveries and security compliance.

Consultant, Lead Software Developer **Sep 2017 – Sep 2021**
Rythmos, client: Unity Technologies | Hyderabad

- Fixed bugs and added features in the **Unity Editor**, including Presets and Enum Flags handling, and worked on **UNET**, Unity's multiplayer networking module.
- Maintained Unity's fork of Uncrustify (open-source code formatter): **fixed 50+ issues**, rebased the fork onto upstream, and contributed fixes that were merged into the upstream project.
- Handled sustained engineering for enterprise customers: root cause analysis, critical fixes and back-ports across C, C++ and C# code.
- Trained **30+ interns** in C and C++. Received the **Mountain Mover Award (Q4 2019)**.

Lead Developer **Dec 2014 – Aug 2017**
NCR Corporation | Hyderabad

- Customized NCR's **Self-Service Checkout (SSCO)** platform for large retail clients, including **Woolworths Australia** and the **US Defense Commissary Agency (DeCA)**: checkout flows, screens and client rules.
- Worked with cross-functional onsite, client and QA teams on requirements and defects; received the **Star of the Sprint** award several times.

Senior Software Engineer **Jan 2010 – Dec 2014**
Aricent (now Capgemini Engineering), client: Alcatel-Lucent | Bangalore

- Built control-plane and user-plane features for Alcatel-Lucent radio network controllers in an Agile team, from coding through integration testing and RCA.
- Developed **TGen**, a traffic generator that simulated NodeBs, user devices and core network behaviour for load and capacity testing of the RNC 9370 and the cloud-based Light Radio Controller.
- Mentored junior developers and ran **advanced C++ training** for employees. Started on Java, JSP and Servlets building an XML-driven UI generator.

PATENTS

7 applications: 3 published, 3 filed, 1 in internal review.

- **Published:** Systems and Methods for Detecting Altered Documents (US20250286723A1, US20250285461A1, US20250286722A1)
- **Filed or in review:** signature forgery detection; contextual document insights through gesture-based interaction; duplicate content identification and regeneration using AI; ownership management of soft copies

INDEPENDENT AI WORK

- Built and trained a **small transformer-based language model from scratch**, then fine-tuned and evaluated it.
- Built **agentic apps** with Pydantic AI, MCP, RAG and A2A (tools, guardrails, memory, structured output): a customer-care bot, Q&A bot, SQL query generator and code-generation agent, containerized with Docker. Code on GitHub.

EDUCATION

Master of Computer Applications (MCA), Sri Venkateswara University, Tirupati · 81.64% **2010**
B.Sc. Mathematics, Statistics and Computer Science, Acharya Nagarjuna University · 79.52% **2007**
Taught C, C++, data structures and algorithms, C# and Java to **500+ students** as part-time faculty while completing the MCA.